

TrialGuard: Market Research and Competitive Positioning

Agentic Point-of-Care Validation for Clinical Trials

Team Ctrl+Agen — KADO Hackathon

1. PROBLEM AND MARKET OPPORTUNITY

Clinical trials are a large, increasingly technology-enabled market. Grand View Research estimates the global clinical trials market at \$89.0B in 2025, with North America accounting for 50.7% of revenue; the market is projected to reach \$158.4B by 2033. Within it, SkyQuest estimates the global *AI in Clinical Trials* market at \$1.69B in 2025, rising to \$4.82B by 2033 at roughly 14% CAGR [1, 2]. This creates a credible wedge for software that improves trial execution rather than only drug discovery or recruitment.

The operational problem is specific. During an ongoing trial, a researcher may want to change a participant’s dose or dosing frequency. That decision must simultaneously satisfy the trial protocol, remain appropriate for the individual patient, and fit the study’s financial/billing rules. Today those checks are commonly fragmented across protocols, EHR/EDC data, clinical decision-support alerts, investigators, monitors, and spreadsheets. This fragmentation matters because protocol deviations are frequent: a Tufts CSDD working-group study of 187 protocols reported a mean of 119 deviations in Phase III and deviations affecting nearly one-third of enrolled patients [3]. FDA guidance also emphasizes that protocol deviations need consistent identification, classification, reporting, and documentation [4].

The accountability problem is visible in adjacent clinical decision-support systems. A 23-study systematic review found medication-alert override rates ranging from 46.2% to 96.2% [5]. In a 2020 tertiary-hospital study in Medina, Saudi Arabia, 6,551 of 7,098 drug–drug-interaction alerts (92.2%) were overridden; 13.0% of those overrides had no reason selected [6].

2. WHERE THE GAP IS

The market already contains strong tools for individual pieces of the workflow. Risk-based monitoring (RBM) is an established regulatory direction: FDA describes monitoring as a quality-control activity intended to determine whether investigation activities are being carried out as planned, while ICH E6(R3) requires proportionate risk identification and control throughout trial conduct and emphasizes documentation and fit-for-purpose computerized systems [8, 9]. Yet adoption remains uneven. A national Australian survey found 55% of responding studies used source-data verification, compared with only 10–11% using risk-based targeted/triggered monitoring [7].

This leaves a narrower product gap: **point-of-care, per-decision validation**. Existing risk frameworks generally operate at study/site level, while traditional alerts generally address one clinical domain. TrialGuard targets the moment a researcher proposes a concrete action and answers three linked questions before that action is taken: *Is it safe for this patient? Is it compliant with this protocol? What is its financial impact?* It then records the human response rather than treating an override as a disappearing alert interaction.

3. TRIALGUARD SOLUTION AND MARKET FIT

TrialGuard is an agentic decision-support layer positioned between trial data and the researcher’s decision. The researcher uses a web interface to enter all the required details. Then the request is passed on to three specialist agents: **Patient Safety, Protocol Compliance, and Financial Risk**. Their outputs are merged into a concise recommendation with grounded evidence, followed by a Human-in-the-Loop (HITL) choice: **Accept, Reject, or Modify**. The choice and recommendation are stored in a database.

The fit with the market is deliberately focused rather than attempting to replace an entire CTMS, EDC, or RBQM platform. TrialGuard can consume existing FHIR/EHR and protocol sources while adding a decision-time validation layer. Its architecture also aligns with current GCP guidelines: ICH E6(R3) specifically calls for attention to risks affecting participant safety and data reliability throughout trial conduct and highlights validation of systems used for critical functions such as dosing and dose titration/reduction [9]. The product thesis is therefore not “AI replaces the investigator”; it is “AI assembles the relevant evidence before the investigator acts, and preserves the resulting decision trail.”

4. COMPETITIVE LANDSCAPE AND DIFFERENTIATION

TrialGuard does not compete on breadth with every clinical-trial platform. Its differentiation is the combination of *protocol compliance + patient-specific safety + financial impact + per-action audit trail* in one point-of-care workflow. Table 1 compares TrialGuard with its market alternatives.

TransCelerate’s RACT is explicitly designed to identify risks affecting subject safety, data quality, and regulatory compliance and to guide monitoring/mitigation plans [10]. CluePoints currently markets centralized, continuous monitoring across clinical, operational, safety, and laboratory data and reports more than 18,900 platform users, 2,800+ studies de-risked, and 320,000+ potential issues detected [11]. These capabilities validate that the underlying market is real, but they also clarify the whitespace TrialGuard is targeting: a lightweight action-level layer rather than another portfolio-level analytics suite.

5. CONCLUSION

TrialGuard addresses a focused, measurable gap inside a rapidly expanding market: validating a proposed trial action before execution across three dimensions that are normally fragmented. The opportunity is supported by the size of clinical-trial and AI-in-trials markets, the documented frequency of protocol deviations, uneven adoption of modern monitoring, and the high override rates observed in adjacent clinical decision-support workflows. Its competitive position is not based on replacing established RBQM or clinical platforms; it is based on adding a point-of-care, protocol-grounded, patient-specific, financially aware decision layer with a persistent human-decision record.

Table 1: TrialGuard vs Market Alternatives

Alternative	Primary focus	Timing	Key limitation for dose-change validation	TrialGuard differentiation
Hospital CDS alerts	Medication safety / interactions	Point-of-care	Usually generic clinical rules; high override rates are documented	Adds trial-specific protocol and financial context plus attributable decision logging.
TransCelerate RACT	Study/protocol risk assessment	Planning / risk management	Framework for categorizing and mitigating risks, not a live per-action dose validator	Converts risk thinking into an execution-time check tied to a specific proposed action.
CluePoints RBQM	Central statistical monitoring, site risk, data quality	Continuous / portfolio	Optimized for detecting statistical and operational risk signals rather than validating one proposed clinical action against protocol text	Uses protocol-aware retrieval and patient-level inputs to produce an action-specific recommendation.
Manual CRC/PI review	Human clinical + protocol judgment	Per decision, but slow	Requires cross-referencing multiple sources and creates variable documentation	Automates evidence gathering while keeping the researcher as final decision-maker.

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